

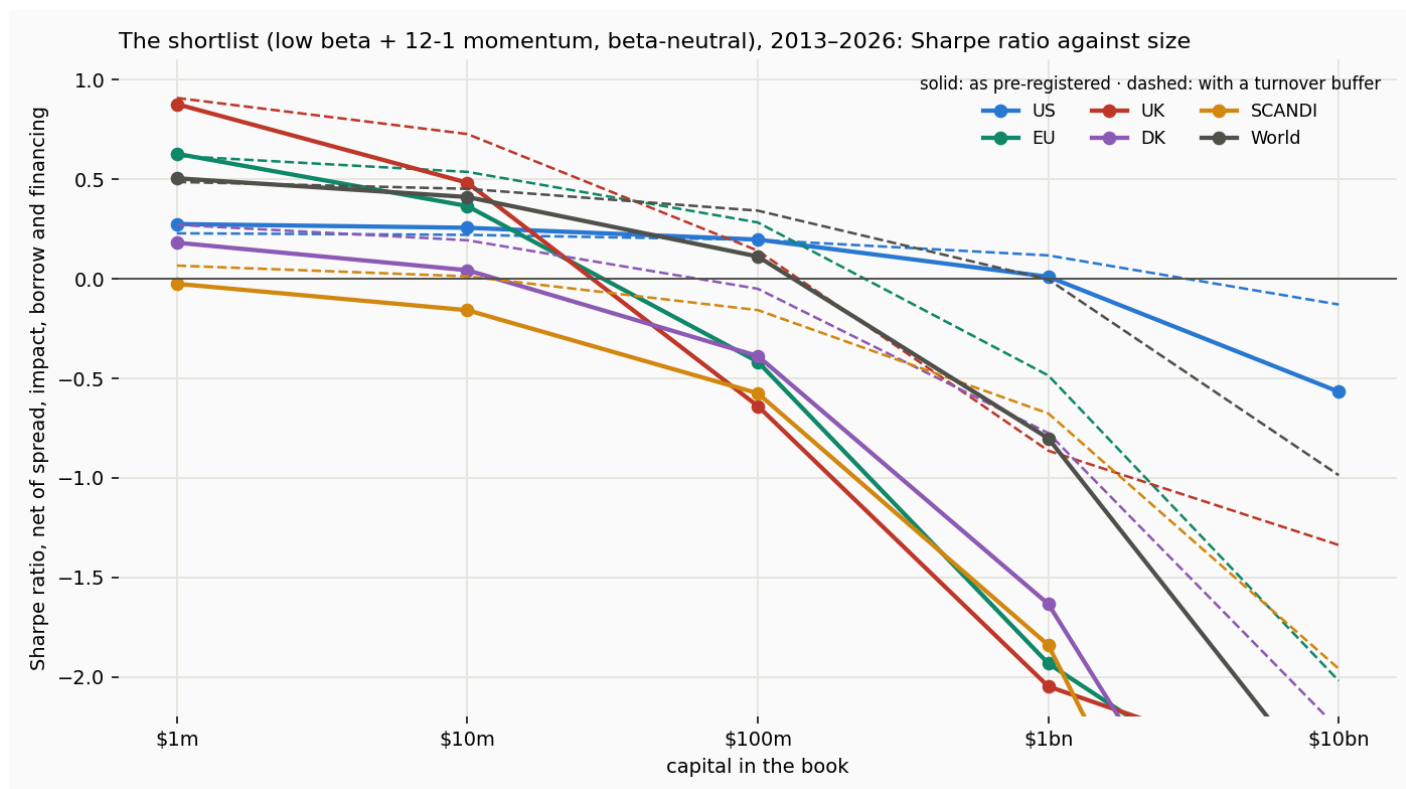
FACTSHEET FS16 · Costs that grow with size

How much money can these strategies hold before trading costs eat them?

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Key facts	
Rules	planning/PREREG_FS16.md, written in the build environment before the run (10:13) and copied to the Factsheets folder right after it (the run finished 10:16); no changes
Cost per trade	half-spread $5 \text{ bp} \times (\text{ADV} / \$50\text{m})^{-0.3}$, 2–50 bp, plus square-root impact $0.7 \times \text{daily volatility} \times \sqrt{(\text{trade} / \text{ADV})}$; sensitivity 1.0
ADV	63-day average traded value in USD per stock (US: price $\times$ volume where the Project1 cache has it; otherwise market cap $\times$ the size-quintile median ratio)
Sizes	\$1m, \$10m, \$100m, \$1bn, \$10bn of capital per book
Books	the shortlist (low beta + 12-1 momentum, beta-neutral) and four raw long/short books, six markets, 2013–2026; borrow fees and financing as before
Replaces	the flat 10 bp per side used in FS01–FS15

In one paragraph. Capacity is the binding constraint for these books, far more than the flat-cost results suggested. The shortlist keeps a positive Sharpe ratio up to about **\$1.0bn in the US** and **\$133m across World**, but only **\$29m in the EU**, **\$27m in the UK** and **\$13m in Denmark**, and nothing in Scandinavia. The UK and EU books look best at small size because their low-beta legs sit in thinly traded stocks, which is also why they fill up first. The fast signals are uninvestable: short-term reversal has a Sharpe ratio of -1.76 to -0.32 already at \$10m. Two exploratory fixes help: a turnover buffer (hold a stock until it leaves the top 30%) halves turnover and raises the size at which the Sharpe ratio halves 3–7 times with little loss at small size; a liquidity screen raises capacity in the EU and World but removes most of the UK result.



1. The shortlist, as pre-registered

Market	\$1m	\$10m	\$100m	\$1bn	\$10bn	Sharpe halves at	Net return reaches zero at
US	+0.28	+0.26	+0.20	+0.01	-0.57	\$208m	\$1.0bn
EU	+0.63	+0.37	-0.42	-1.93	-2.89	\$12m	\$29m
UK	+0.88	+0.48	-0.64	-2.05	-2.55	\$11m	\$27m
DK	+0.18	+0.04	-0.39	-1.63	-3.97	\$5m	\$13m
SCANDI	-0.02	-0.16	-0.57	-1.84	-4.74	—	—
World	+0.50	+0.41	+0.11	-0.80	-2.66	\$34m	\$133m

Sharpe ratio, net of spread, impact ($Y = 0.7$), borrow fees and financing, 2013–2026. Capital per book; the combined shortlist puts half in each leg.

2. Every book, every market

Book	Market	Flat 10 bp	\$1m	\$100m	\$1bn	\$100m, $Y = 1.0$	Spread cost / yr	All-in cost / yr at \$100m	Monthly turnover	Net return reaches zero at
low beta (beta-neutral)	US	+0.13	+0.15	+0.11	+0.01	+0.09	0.3%	1.9%	0.42	\$1.1bn
	EU	+0.56	+0.47	-0.30	-1.36	-0.60	0.9%	27.5%	0.45	\$30m

Book	Market	Flat 10 bp	\$1m	\$100m	\$1bn	\$100m, Y = 1.0	Spread cost / yr	All-in cost / yr at \$100m	Monthly turnover	Net return reaches zero at
	UK	+0.98	+0.84	-0.39	-1.49	-0.79	1.4%	40.1%	0.50	\$38m
	DK	+0.08	+0.06	-0.22	-0.85	-0.35	0.3%	5.7%	0.20	\$7m
	SCANDI	+0.09	+0.08	-0.18	-0.77	-0.30	0.4%	5.2%	0.27	\$12m
	World	+0.31	+0.30	+0.02	-0.62	-0.11	0.7%	9.9%	0.48	\$107m
12-1 momentum (beta-neutral)	US	+0.29	+0.33	+0.23	-0.00	+0.19	0.5%	2.3%	0.57	\$975m
	EU	+0.55	+0.48	-0.36	-1.90	-0.73	1.0%	16.3%	0.57	\$27m
	UK	+0.70	+0.54	-0.74	-2.23	-1.20	1.3%	28.8%	0.55	\$16m
	DK	+0.26	+0.22	-0.37	-1.68	-0.64	0.7%	11.7%	0.43	\$15m
	SCANDI	-0.10	-0.14	-0.75	-2.14	-1.04	0.8%	10.2%	0.54	—
	World	+0.55	+0.55	+0.22	-0.56	+0.06	0.8%	6.3%	0.56	\$189m
12-1 momentum (L/S)	US	+0.32	+0.36	+0.27	+0.05	+0.22	0.5%	2.5%	0.58	\$1.2bn
	EU	+0.34	+0.28	-0.42	-1.72	-0.73	0.9%	17.0%	0.53	\$16m
	UK	+0.43	+0.30	-0.74	-2.17	-1.14	1.3%	29.1%	0.53	\$11m
	DK	+0.31	+0.27	-0.26	-1.43	-0.50	0.6%	11.3%	0.38	\$22m
	SCANDI	-0.19	-0.22	-0.75	-1.98	-1.00	0.7%	9.8%	0.48	—
	World	+0.41	+0.41	+0.11	-0.58	-0.03	0.8%	6.9%	0.56	\$145m
low volatility (L/S)	US	-0.60	-0.59	-0.61	-0.67	-0.63	0.2%	0.9%	0.22	—
	EU	-0.08	-0.09	-0.28	-0.71	-0.37	0.3%	4.6%	0.20	—
	UK	-0.30	-0.35	-0.74	-1.26	-0.89	0.5%	10.9%	0.18	—
	DK	-0.08	-0.10	-0.32	-0.82	-0.42	0.3%	5.0%	0.16	—
	SCANDI	-0.26	-0.27	-0.45	-0.86	-0.53	0.3%	3.5%	0.18	—
	World	-0.48	-0.48	-0.58	-0.81	-0.63	0.3%	2.5%	0.20	—
short-term reversal (L/S)	US	-0.36	-0.24	-0.57	-1.35	-0.72	1.4%	7.9%	1.74	—
	EU	-0.61	-0.83	-3.18	-5.74	-3.98	2.8%	53.0%	1.72	—
	UK	-0.17	-0.67	-4.46	-6.51	-5.39	4.1%	91.4%	1.71	—
	DK	-0.38	-0.58	-2.98	-6.98	-4.00	2.1%	46.0%	1.34	—
	SCANDI	+0.01	-0.14	-2.38	-6.72	-3.37	2.4%	36.6%	1.58	—
	World	-0.49	-0.51	-1.64	-4.20	-2.18	2.4%	21.9%	1.73	—
low MAX (L/S)	US	-0.81	-0.71	-0.95	-1.52	-1.06	1.2%	6.1%	1.51	—
	EU	-0.09	-0.27	-2.42	-5.47	-3.28	2.5%	40.1%	1.51	—

Book	Market	Flat 10 bp	\$1m	\$100m	\$1bn	\$100m, Y = 1.0	Spread cost / yr	All-in cost / yr at \$100m	Monthly turnover	Net return reaches zero at
	UK	-0.20	-0.70	-3.12	-3.96	-3.50	3.7%	81.2%	1.40	—
	DK	-0.26	-0.41	-2.12	-5.41	-2.87	1.7%	35.0%	1.09	—
	SCANDI	-0.17	-0.28	-2.04	-5.76	-2.84	2.1%	28.8%	1.38	—
	World	-0.69	-0.70	-1.66	-3.68	-2.10	2.2%	18.1%	1.50	—

Reading it.

1. **At \$1m the model is close to the flat 10 bp in large-cap markets and harsher in small ones:** the spread alone costs more than 10 bp per side in the UK, EU and Scandinavian small caps.
2. **Impact grows with the square root of size,** so costs rise about three times for every tenfold increase in capital, and far faster in markets where the traded stocks are thin.
3. **Slow signals scale; fast ones do not.** Low volatility and low beta change little from month to month; reversal and low MAX replace most of the book every month and collapse even at \$10m.
4. **Beta-neutral books pay twice:** their low-beta leg is levered up (about 2× in most markets), so its trades are twice as large as the capital suggests.

3. Exploratory fix 1 · A turnover buffer (not pre-registered)

Market	Monthly turnover (was about 0.5)	\$1m	\$10m	\$100m	\$1bn	Sharpe halves at	Net return reaches zero at
US	0.22	+0.23	+0.22	+0.20	+0.12	\$1.0bn	\$3.0bn
EU	0.23	+0.62	+0.54	+0.28	-0.49	\$79m	\$233m
UK	0.24	+0.91	+0.73	+0.14	-0.86	\$29m	\$138m
DK	0.20	+0.27	+0.19	-0.05	-0.77	\$17m	\$62m
SCANDI	0.18	+0.07	+0.01	-0.16	-0.68	\$4m	\$12m
World	0.23	+0.49	+0.45	+0.34	-0.00	\$193m	\$978m

The FS03–FS12 fix ladder's buffer: a stock enters in the top decile (quintile, tercile) and is sold only when it leaves the top 30%.

4. Exploratory fix 2 · A liquidity screen (not pre-registered)

Market	Stocks left (median)	\$1m	\$10m	\$100m	\$1bn	Sharpe halves at	Net return reaches zero at
US	652	+0.27	+0.26	+0.20	+0.01	\$208m	\$1.0bn
EU	298	+0.45	+0.37	+0.12	-0.67	\$38m	\$141m
UK	149	+0.24	+0.10	-0.37	-1.83	\$7m	\$16m

Market	Stocks left (median)	\$1m	\$10m	\$100m	\$1bn	Sharpe halves at	Net return reaches zero at
DK	24	+0.25	+0.11	-0.33	-1.60	\$8m	\$18m
SCANDI	79	+0.00	-0.12	-0.52	-1.75	\$1m	\$1m
World	1095	+0.36	+0.33	+0.24	-0.06	\$155m	\$627m

Stocks with less than \$5m average daily traded value removed before sorting.

5. What it means

1. **The factsheet results are small-fund results.** Outside the US and World, the shortlist's edge disappears somewhere between \$10m and \$100m per book with monthly rebalancing.
2. **A buffer is the cheapest capacity there is:** it keeps most of the small-size Sharpe ratio and multiplies capacity.
3. **The UK low-beta premium lives in thin stocks.** Screened for liquidity, it largely goes away: a warning for anyone reading FS10's UK result as investable at scale.
4. **For the multifactor model,** cost-aware construction (buffers, trading towards targets over several days, liquidity-weighted positions) matters more than the choice of signals.

6. Caveats

- One cost model with textbook parameters; real costs depend on execution skill, venue and timing. $Y = 1.0$ is shown as the harsher case.
- US volume data are missing for about a third of the stocks ever in the top 500 (mostly delisted names); their ADV comes from market cap, a proxy.
- Trades are costed as if done in one day; spreading large trades over several days lowers impact but adds tracking error.
- The exploratory sections were decided after seeing the pre-registered results and are not tests.

7. References

- Almgren, R., Thum, C., Hauptmann, E. & Li, H. (2005). Direct estimation of equity market impact. *Risk* 18(7), 58–62.
- Frazzini, A., Israel, R. & Moskowitz, T. J. (2018). Trading costs. Working paper, AQR Capital Management.
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- Bouchaud, J.-P., Bonart, J., Donier, J. & Gould, M. (2018). *Trades, Quotes and Prices*. Cambridge University Press.

8. Reproduce

planning/PREREG_FS16.md → code/impact.py (run, and the exploratory run_buffer, run_liquid) → code/build_fs16.py. US volume from the Project1 price cache (_data_private/pit/US_tradedvalue.parquet, private).